Project Proposal

Contents

Research – 4-13

Set task brief

You work for a software development company. The company has secured a new contract to develop a digital solution for Greenfield Local Hum (GLH). GLH is a cooperative of local farmers and food producers. GLH offers:

- A single location retail outlet for locally produced food and drink
- Collective marketing for local farmers and food producers
- Product management, stock control, and product traceability

GLH (the client) would like to develop a digital solution that will:

- Provide information about the producers, their methods, and the benefit of buying locally
- Allow customers to:
    - View products with transparent pricing and availability
    - Place orders for collection or delivery
    - Manage their accounts and view order history
- Provide a dashboard to enable producers to manage stock levels, update product details, and view orders

The client has conducted market research with existing customers to identify features that could be included in the digital solution. The features suggested by the client are:

- Order management, scheduling and tracking features
- A loyalty scheme that provides discounts and special offers
- Accessibility features to support a wide range of users.

Sections

- Business context
- Project aim
- Functional Requirements
- Non-Functional Requirements
- Hierarchy Diagram
- UI considerations
- Security and Risk mitigation (risk matrix)
- Legal and regulatory guidelines

Above is a reminder for the task and not part of the exam!

Research

Website 1: Riverford.co.uk

Summary

Riverford is an online farm shop that delivers farm grown and organic food to customers while promoting and advertising sustainability and ethical food production.

System Requirements

Since the website is opened on a standard browser, it means the website can be accessed on many devices.

Low hardware is needed, as long as the user has access to a browser and the internet, the user is able to access the website.

It's compatible with many different browsers, such as: Google Chrome, Microsoft Edge, and Safari. Due to this, this makes the website widely accessible and available.

Overview of the features

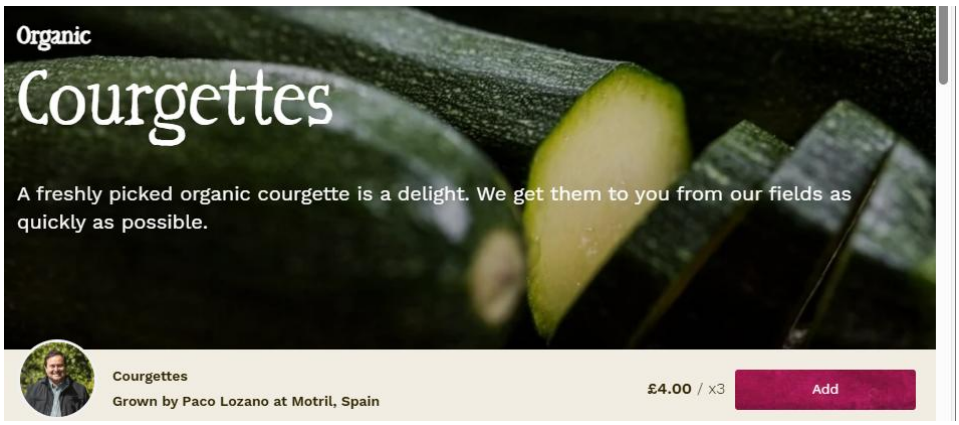
The target audience is people who are interested in organic food and want produce that is locally sourced. Due to Riverford promoting sustainability and ethical food production, the target audience also will be people who care about the environment and want to stay sustainable. Families who also want fresh food delivered to their home will also be the target audience!

Generic features include:

- A product catalogue showing available fruits and vegetables
- Pricing and product information
- A basket and checkout
- Customer accounts
- Delivery schedules
- Order History

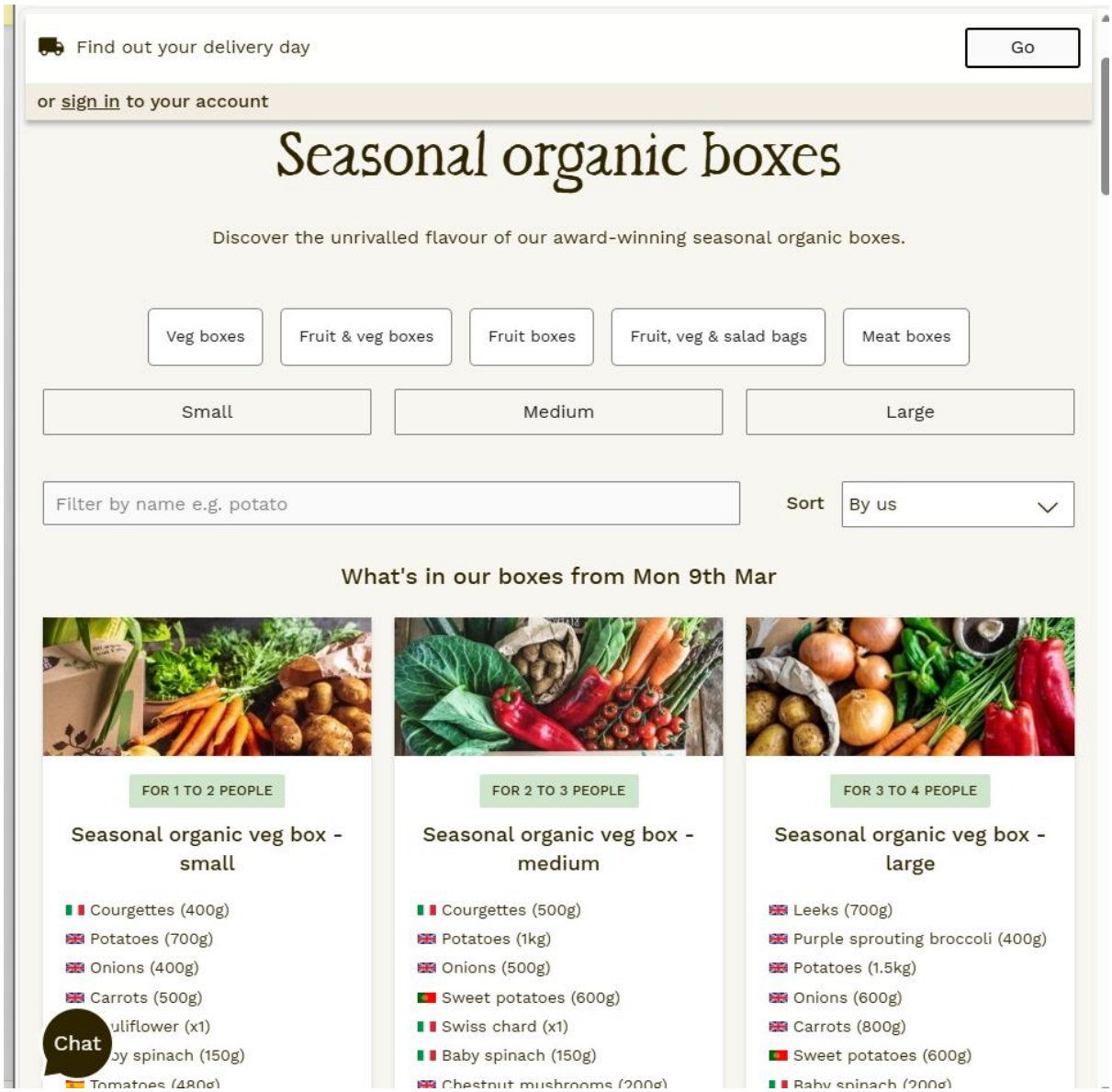
All these features are industry standard for a shopping website

Unique Features



One unique feature of Riverford is that it tells the users where the food was grown and who produced it! Usually on shopping websites, such as Tesco and Asda, you may be able to find where something was produced on a packaging, but definitely not who made it. This uniqueness builds

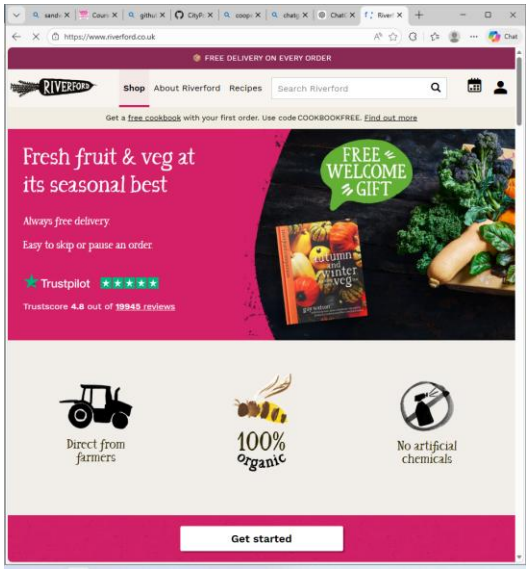
trust with a customer because they are able to know exactly where their food is coming from. This is especially good because Riverford promotes sustainability and ethical food production! A lot of users and customers want to support their local ethical and farming too, so this encourages users to buy from Riverford! However, this requires extra data management and storage, because users who upload their organically grown food will need their food linked to their name and account through the database!

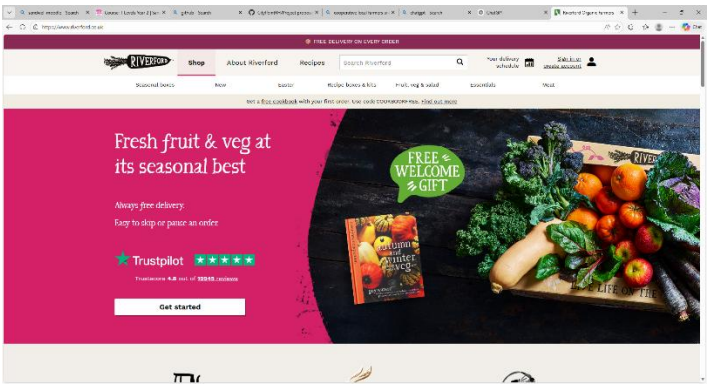


Another unique feature they have is seasonal vegetable boxes! This means that customers are able to purchase a box with seasonal fruits and vegetables. This proves to the users that the company promotes sustainable farming because crops are being harvested at their natural season instead of being genetically modified (as most fruits and vegetables are in supermarket chains) to stay ripe for longer! This also helps reduce food waste, because farmers are only harvesting what's needed. However, these boxes mean that users have less control over each individual item. That is important because if someone was allergic to something in the box, they wouldn't be able to remove said item!

<u>Newly emerging technology</u>

The technology used is touch-based interfaces for users who are viewing the website on mobile, responsive design for different screen sizes, (if you were to decrease the size of the windows, the elements follow suit), and online account systems with automated ordering.





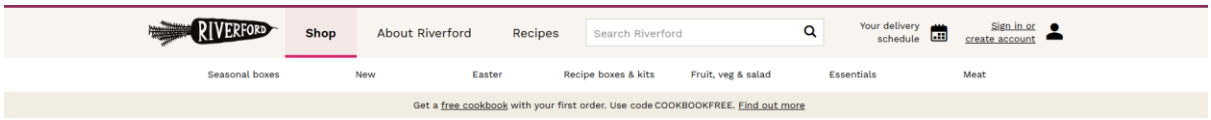
Above is an example of the elements adjusting based on screen size!

The touch-based interface is effective for phones and tablets, this makes the website more accessible because many users have mobile devices to carry around instead of desktops or laptops.

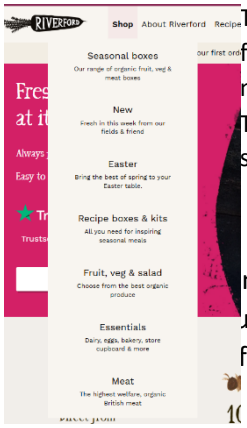
<u>User Interface</u>

<u>*Overall Design*</u>

The interface has cleat navigation menus via a navigation bar at the top of the screen

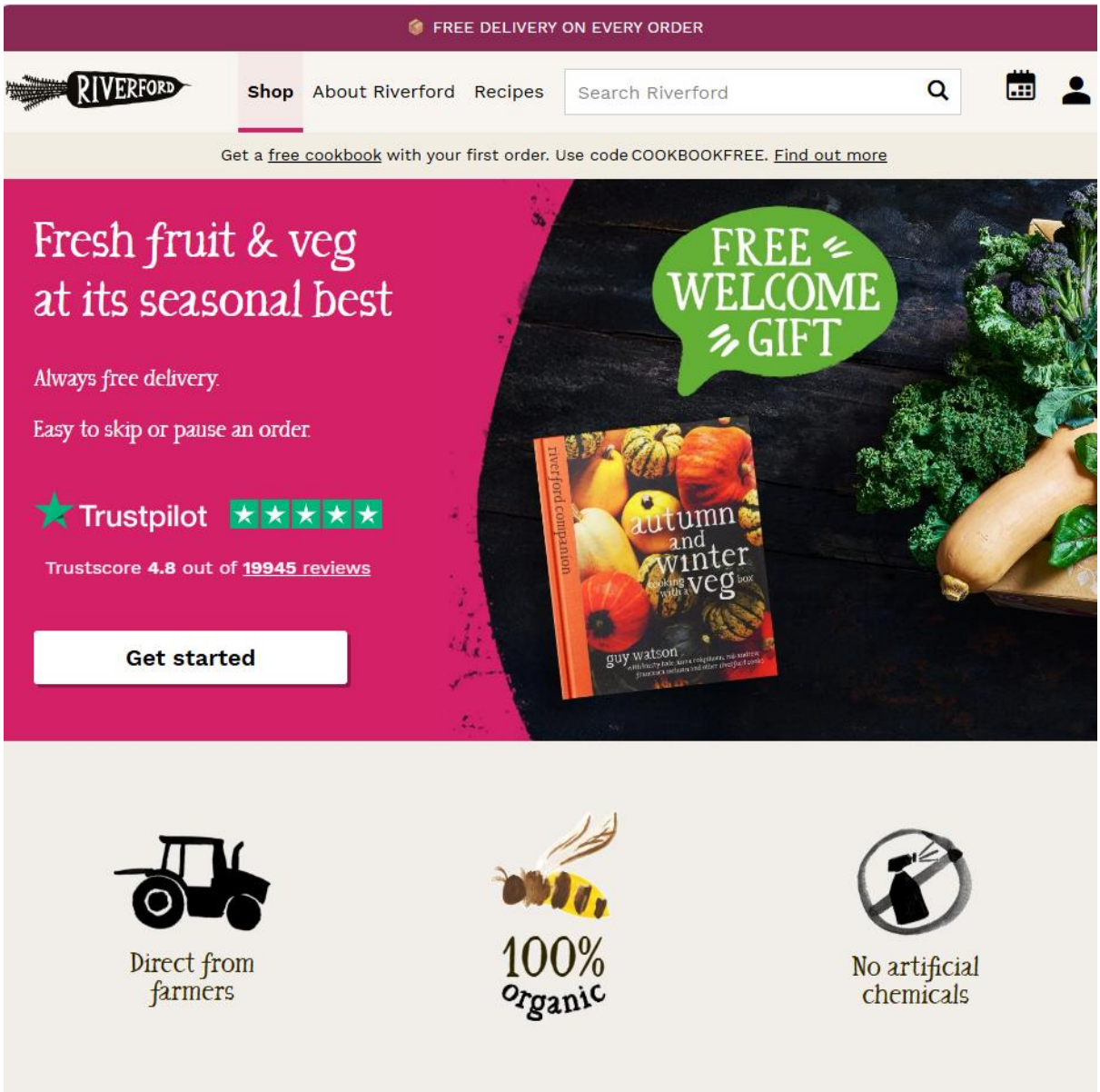


And when the screen size is adjusted it turns the nav par into a drop down!



This makes the navigation especially easy to understand, mostly because this feature is industry standard, many websites have their nav bar set out like this, making this website accessible to even people with low technical capabilities. This makes important information especially easy to find! And makes the system suitable for a wide range of users

ns, the colours used are mostly green, pink, and an off-white. Green is usually ure, farming, and sustainability, and the pink contrasts it in a nice way. It almost farming itself. Which supports Riverford's brand identity



Graphics include high quality photos of vegetable and food, photos of the farmers themselves, and the product images.

From what I have found, there are no videos or animation. This helps the website load quickly and not overwhelm the user with too much information

The fonts are a simple sans-serif font and Times New Roman for headings. Serif fonts usually improve readability for users and make the website feel professional and trustworthy! This is usually industry standard, as any serif fonts are usually considered accessible when it comes to reading disabilities

We offer a variety of set boxes to suit your needs, harvested from our own fields and grower friends. Each box is carefully picked and packed by us every week.

Choose a weekly box of goodness

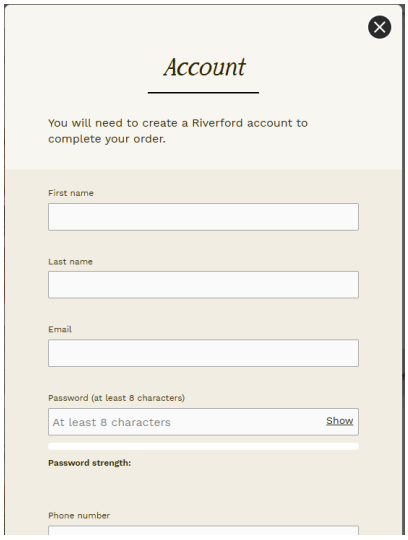
The language used is friendly, informative, and easy to understand. It doesn't use any language that might be too technical for an average user who isn't a farmer! This makes the website accessible to a wide range of users including their target audience as you can see above!

Accessibility Features

From what I can tell, Riverford does not have an accessibility menu, therefore more advanced features, such as: Text-to-speech, adjustable font sizes, or dark mode are unavailable therefore it narrows down the customer base that Riverford could be reaching out to. However, it does have clear navigation, readable fonts, responsive design, FAQ and Live chat. These features help a wide range of users to access the website!

Security and Legal

The website includes secure login accounts, password authentication, HTTPS encryption and cookie notification. These help protect user data and information while also ensuring no unauthorised access is allowed to take a user's private information!



The website ensures you need to make an account in order to check out from the basket!

The website also follows legal standards as the General Data Protection Regulation (GDPR) that protects personal data collected from users. It follows the standard by making the user accept cookies before data is collected, personal data is stored securely, and users can access privacy policies explaining how their data is being used.

Another law is the Data Protection Act 2018 (DPA) which works alongside GDPR to control how businesses use and handle personal data. The website follows DPA by storing customer account information securely, allowing users to request deletion or access to their data, and explaining how information is collected and used

Consumer Rights Act 2015 (CRA) protects customers when buying goods or services online. The website follows this law by providing accurate product descriptions, clearly showing pricing, and allowing users to complain or request refunds if the product is faulty.

Riverford also follows the copyright, designs, and patents act. This law protects images, videos, text, and other creative content. They follow this law by using original photos of farms and produce, it owns or has permission to use website images and graphics, and it protects its own content from being copied without permission.

Website 2: Abel & Cole

Summary

Abel & Cole is an online shopping retail platform that delivers sustainable and organic food directly to customers! The website promotes and advertises organic farming, ethical food, and environmentally friendly ways to manage food production!

System Requirements

Since the website is opened on a standard browser, it means the website can be accessed on many devices.

Low hardware is needed, as long as the user has access to a browser and the internet, the user is able to access the website.

It’s compatible with many different browsers, such as: Google Chrome, Microsoft Edge, and Safari. Due to this, this makes the website widely accessible and available.

The website is also mobile responsive meaning it adjusts to different screen sizes!

Overview of Features

The target audience of Abel & Cole is very similar to Riverford.

The target audience includes: Customers who are interested in organic food, people who care about sustainability and ethical farming, families who want fresh food delivered to their home, and environmentally conscious consumers who want to support local producers

Generic features include:

- A product catalogue showing available fruits and vegetables
- Pricing and product information
- A basket and checkout
- Customer accounts
- Delivery schedules
- Order History

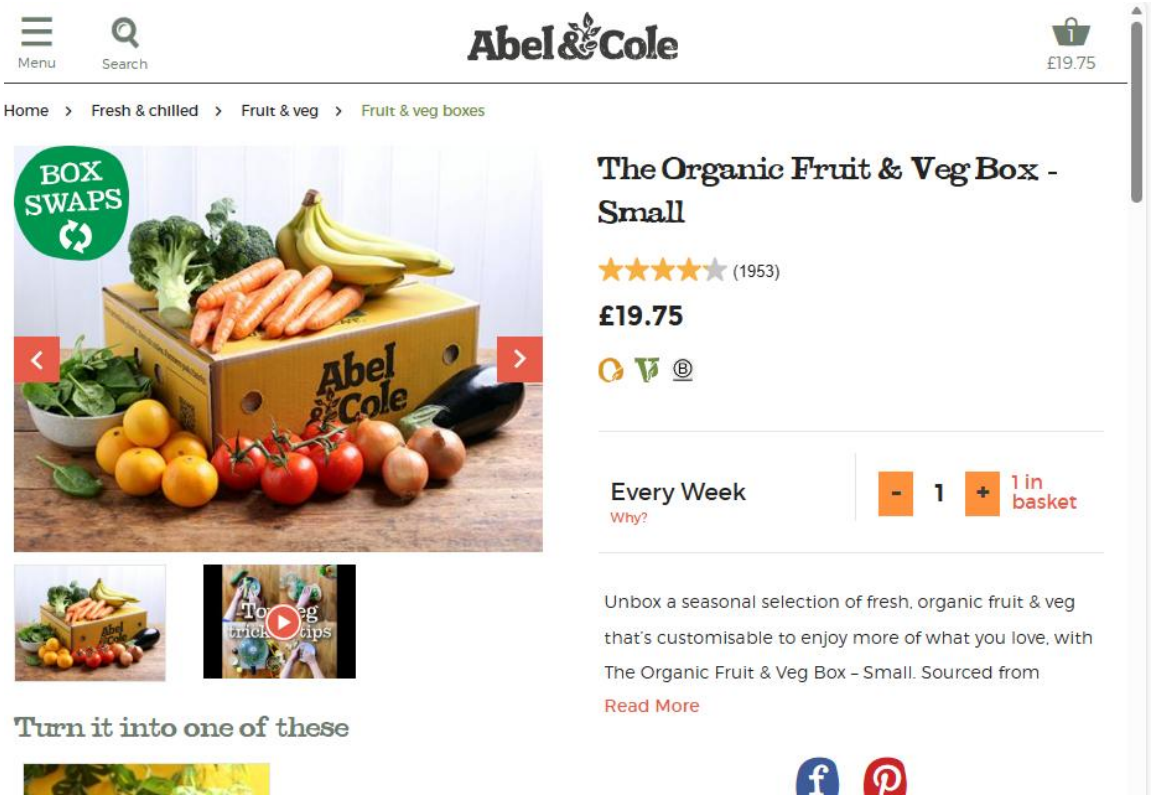
All these features are industry standard for a shopping website

Unique Features

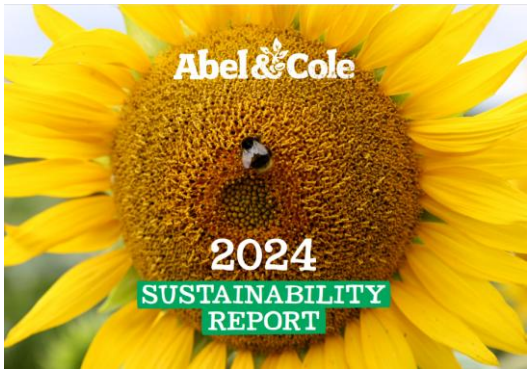
One unique feature of the Abel & Cole website is their organic food subscription boxes. Customers are able to: Subscribe to weekly fruit and vegetable boxes, choose the size of the box, and select delivery days. This allows customers to receive regular deliveries of fresh organic produce without needing to manually place an order each week.

This feature is good because it makes shopping more convenient for busy customers. It also helps the company predict demand and reduce food waste because farmers know how much produce needs to be harvested.

However, this feature also has some disadvantages. Because the boxes are pre-selected, customers may have less control over the specific items they receive. This could be bad if customers dislike certain foods or have allergies.

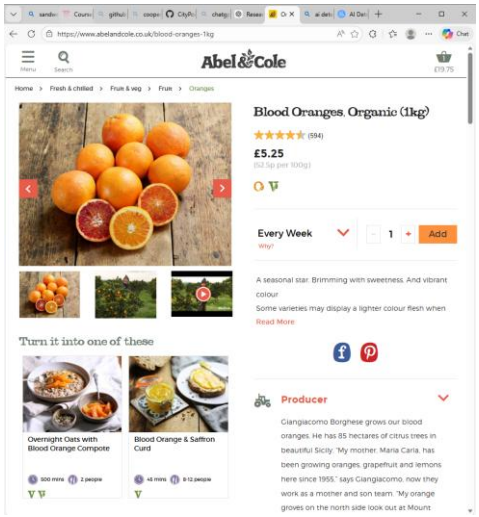


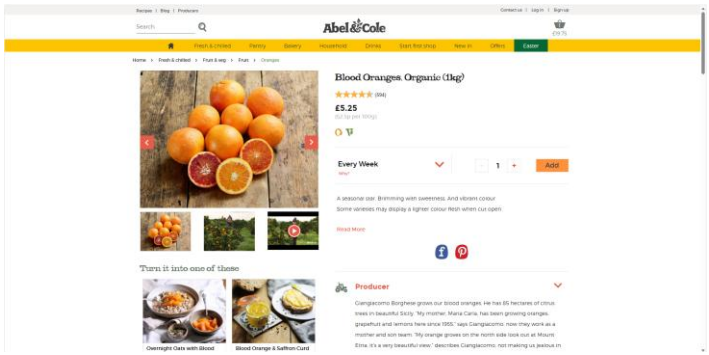
Another unique feature is sustainability tracking. The website emphasises the importance of environmental practices, such as: Low-carbon farming, reduced packaging, and organic farming standards. This feature appeals to environmentally friendly people and encourages customers to engage in such practices



<u>Newly Emerging Technologies</u>

The technology used is touch-based interfaces for users who are viewing the website on mobile, responsive design for different screen sizes, (if you were to decrease the size of the windows, the elements follow suit), and online account systems with automated ordering.





Above is an example of the website being fully open vs resized.

Overall Design

Overall, the design is simple, modern and visually appealing. The navigation bar is at the top of the website, and when the screen is resized the navigation bar turns into a hamburger menu as seen in the images above! This improves usability on mobile devices! The layout overall is industry standard for a shopping website.

The colours used are mainly white, amber, and orange. This allows for contrast for the website whilst keeping a clean modern look

The graphics uses are high quality images of produce and food photography, whilst also using icons to make the website more visually accessible. This helps the users engage with the website while being able to see what they're ordering.

The fonts used are ACMontserratLightFont, ACFont, Which aren't considered industry standard, as serif fonts are. This is because serif fonts allow easier readability! However, the text is big enough for any user to read!

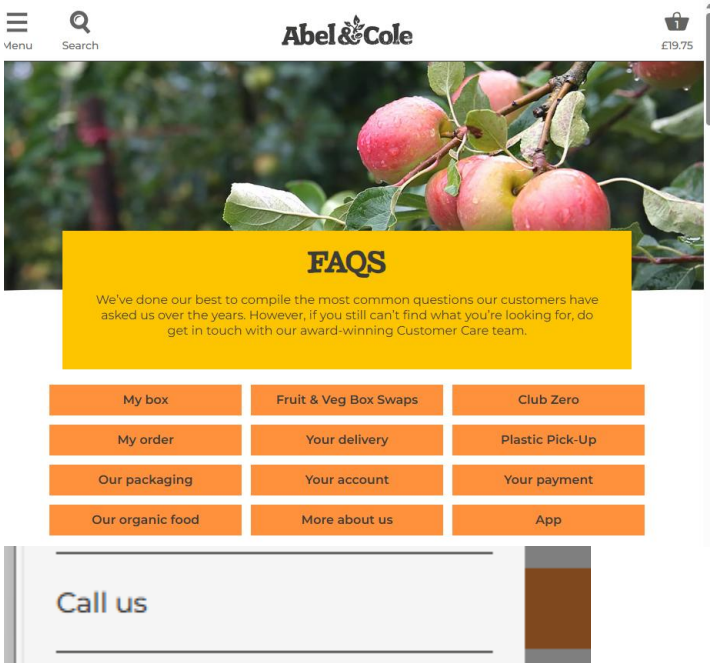
Giangiacomo Borghese grows our blood oranges. He has 85 hectares of citrus trees in beautiful Sicily. "My mother, Maria Carla, has been growing oranges, grapefruit and lemons here since 1955," says Giangiacomo, now they work as a mother and son team. "My orange groves on the north side look out at Mount

Blood Oranges, Organic (1kg)

The language they use is a friendly tone with non-technical language so any customer who doesn't know how to farm can understand what they're reading!

Accessibility Features

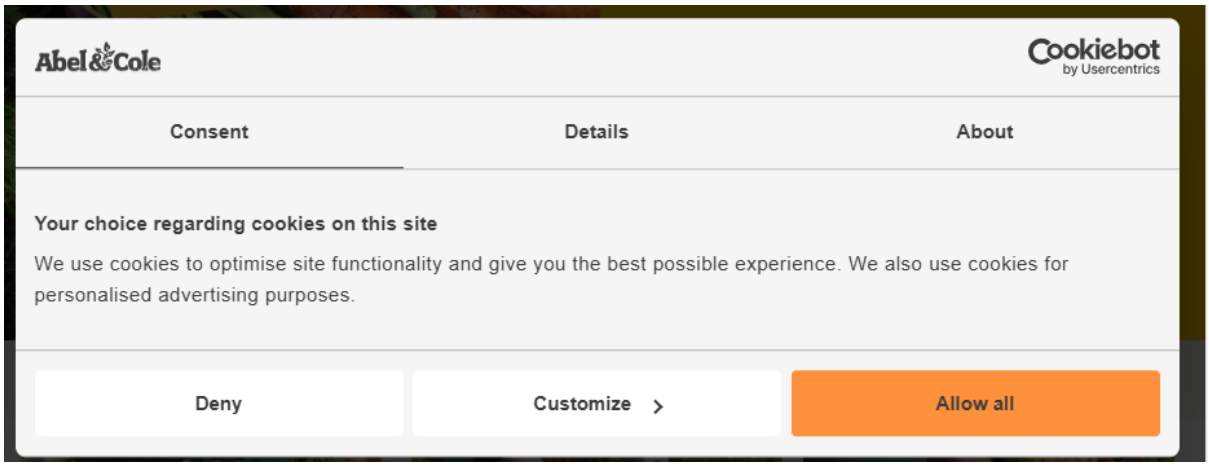
I do not see an accessibility menu, which means that users are unable to change the website to dark mode, or adjust font text, or enable Text-To-Speech. However, the website does have help tools such as an FAQ and a call us section!



However, due to the lack of an accessibility menu, the accessibility is not effective!

Security and legal

The website includes secure login accounts, password authentication, HTTPS encryption and cookie notification. These help protect user data and information while also ensuring no unauthorised access is allowed to take a user's private information!



The website also follows legal standards as the General Data Protection Regulation (GDPR) that protects personal data collected from users. It follows the standard by making the user accept cookies before data is collected, personal data is stored securely, and users can access privacy policies explaining how their data is being used.

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Abel & Cole also follows the copyright, designs, and patents act. This law protects images, videos, text, and other creative content. They follow this law by using original photos of farms and produce, it owns or has permission to use website images and graphics, and it protects its own content from being copied without permission.

Business Context

A single location retail outlet for locally produced food and drink

The client offers a single location retail outlet for locally produced food and drinks! Making the website would allow customers to discover and support Greenfield Local Hub (GLH) and make purchases easier. Especially because a single location is not guaranteed to have many customers, and a lot of people go shopping for groceries online now! However, being a single retail outlet for locally produced foods and drinks shows customers that everything is local and not around the world. This helps increase sails because people will trust GLH to be completely organic and sustainable, and its people who care about the environment who are the target audience.

Collective marketing for local farmers and food producers

The client wants the platform to support collective marketing for local farmers and food producers. By bringing multiple food producers under one platform, the GLH can promote a variety of different crops and produce which will be suitable for any demographic, and be accessible to customers who are picky in taste, or have allergies. This also benefits farmers who don't have the resources to sell independently and attracts customers who want to support their local businesses and are interested in being sustainable and environmentally friendly.

Product management, stock control, and product traceability

The client requires a system for product management, stock control, and product traceability. This ensures that GLH can be accurate in tracking different available products, updating stock levels, and managing the inventory efficiently. Traceability is important in food industries because it allows GLH to identify where the products come from and respond quickly to any quality and safety issues, because if there are any quality issues and GLH doesn't have the traceability to fix it, then GLH will lose money. This improves efficiency and ensures customers get the most accurate information possible.

Provide information about the producers, their methods, and the benefit of buying locally

The client wants to provide information about the producers, their methods, and the benefits of buying locally. Providing information is especially important when promoting organic and a sustainable website, as it builds trust between GLH and their target audience. This helps the audience understand the story behind each produce they buy and create an emotional attachment between the producers and the customers. This means GLH is more likely to have customer retention because of the emotional connection. Displaying their methods and the benefits of buying locally also builds that same trust.

View products with transparent pricing and availability

The client wants customers to be able to view products with transparent pricing and availability. This allows customers to browse products easily and compare items before making a purchase. Transparent pricing and availability help prevent customers getting frustrated by unavailable products and know how much exactly a product is. Transparent availability also means the customer will be able to see when it's next available and wait for that time to purchase the produce!

Place orders for collection or delivery

The client wants the system to allow customers to place orders for collection or delivery. Offering both options provide flexibility and convenience for customers with different needs and preferences. Some customers may prefer collecting orders to save on delivery costs, while others may prefer the convenience of home delivery. This flexibility helps increase customer satisfaction and makes the service accessible to a wider audience. Especially because some people may live 1 hour away from the store, making them be able to choose the option to have delivery will make it more likely people buy from GLH.

Manage their accounts and view order history

The client has asked to include the ability to manage their accounts and view order history. This improves the user experience by allowing customers to quickly reorder items they have purchased before. It also encourages repeat purchases and helps build long-term customer loyalty with the business. And by managing their account, a farmer would be able to upload photos they took themselves about the produce they're selling, or see what people have ordered to test what produce people want more in stock! This helps increase sales as selling produce people want means more purchases.

Dashboard to enable producers to manage stock levels, update product details, and view orders

The client has requested a dashboard that allows producers to manage stock levels, update product details, and view orders. Giving producers direct access to manage their products ensures information stays accurate and up to date, which is important because if the information is inaccurate, customers would get frustrated and customer retention wouldn't stay. Managing stock levels is important with viewing orders because if there's not enough stock for the number of orders, then customers are bound to not get the product they purchased, meaning GLH would be losing money!

Order management, scheduling and tracking features

The client wants the system to allow order management, scheduling, and tracking features. This ensures the orders are processed in the most efficient way possible. Order management enables organisation for GLH, so instead of getting lost with the amount of orders coming through, they'll be able to get managed so no farmer accidentally misses a customers order ensuring customer retention is still happening. Scheduling and would enable customers to track when their order is prepared to be picked up or where it is in its delivery, which increases customer trust with GLH. This also helps the business maintain organisation and ensures the customers receive their order in time.

Loyalty scheme that provides discounts and special offers

The client wants to introduce a loyalty scheme that rewards repeat customers with discounts and special offers. Loyalty schemes encourage customers to continue purchasing from the business instead of competitors. This helps increase long-term revenue and strengthens the relationship

between the business and its customer base because if the user were to purchase from a competitor, they would lose their discounts and offers.

Accessibility features to support a wide range of users.

The client has requested accessibility features to support a wide range of users. These features ensure that the platform can be used by people with different abilities and needs, promoting inclusivity. Accessibility is also an industry standard and helps the business comply with legal accessibility requirements while expanding the target audience. If there were no accessibility features then people who need the features are excluded, which causes a decrease in revenue

Project Aim

The aim of the project is to produce an online web application that has the following features:

- A single location retail outlet for locally produced food and drink
- Collective marketing for local farmers and food producers
- Product management, stock control, and product traceability
- Provide information about the producers, their methods, and the benefit of buying locally
- Allow customers to:
    - View products with transparent pricing and availability
    - Place orders for collection or delivery
    - Manage their accounts and view order history
- Provide a dashboard to enable producers to manage stock levels, update product details, and view orders
- Order management, scheduling and tracking features
- A loyalty scheme that provides discounts and special offers
- Accessibility features to support a wide range of users.
-

Functional Requirements and User Acceptance Criteria

User Stories

Timmy is a farmer, every season he has a bountiful harvest and wants the ability to sell his produce as it's simply too much for him!

Lizzie doesn't know anything about crops, but she's openly environmentally friendly. She doesn't like chemicals in her food, and wants to eat organically and healthily.

Joel is a single dad with 2 kids, he simply doesn't have the time to go out shopping to buy healthy, organic food. But he's interested in it because he wants his kids to grow up healthy.

Functional Requirement	**User Story**	**User Acceptance Criteria**	**Justification**
Register	As a Timmy I want to register So that I can sell my crops As a Lizzie		

	I want to register So I can purchase from my local farmers healthy food!		
Log In	As a Joel I want to log in So that I can browse information about organic food As a Timmy I want to log in So that I can upload my crops to sell them		

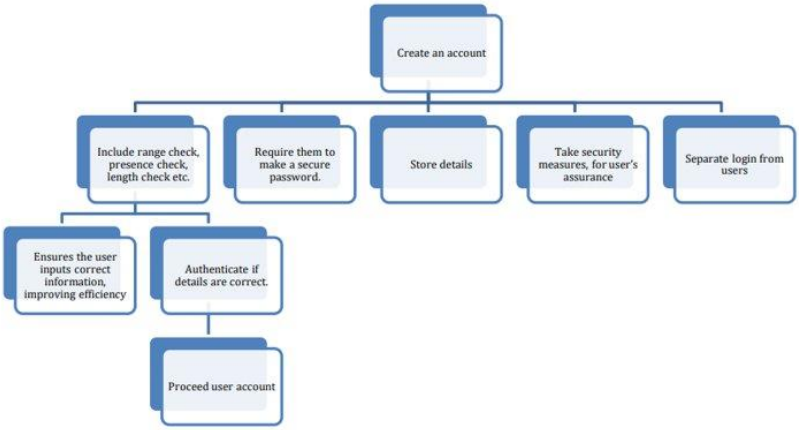
Non-Functional Requirements

Performance		
Non-Functional Requirement	Justification	KPI
Security		
Non-Functional Requirement	Justification	KPI
Scalability		
Non-Functional Requirement	Justification	KPI
Capacity		
Non-Functional Requirement	Justification	KPI
Reliability		
Non-Functional Requirement	Justification	KPI
Usability		
Non-Functional Requirement	Justification	KPI
Accessibility		

Non-Functional Requirement	Justification	KPI

Hierarchy Diagram

- Draw a diagram showing the order in which the functional requirements must be completed.



UI Consideration

- Briefly discuss the considerations that you will have to think about when making the user interface thinking about the target audience.
- Talk about colours, images, layout, text, content and more.

Security Considerations

- What security risks will be involved in creating your web-based solution?
- How will you mitigate these potential risks?
- Make sure each one is specific to your particular solution and is not just generic

Legal and Regulatory Guideline Considerations

- What legal guidelines must you consider when creating the web-based solution?
- Be specific about both computing legislation but also industry related legislation that you must follow